



SUNFLASER
OPTOELECTRONIC SOLUTIONS

SFA3000B

Laser Rangefinder Module



Product Manual

Version 202606V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

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Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- › Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- › Continuing to operate the ranging module after improper handling or misuse;
- › Using a ranging module with obvious faults, defects, or damage;
- › Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- › Eye injury
- › Incorrect results
- › Measurement errors
- › Property damage
- › Equipment malfunction

Operator Responsibilities

- › Be familiar with the basic knowledge required for the correct operation of the ranging module;
- › Understand and comply with the operation and maintenance requirements of the ranging module;
- › Prevent damage to the ranging module during storage and transportation;
- › Keep the ranging module clean and well maintained;
- › Ensure that the ranging module is stored and operated in a safe environment;
- › Do not expose the ranging module to strong mechanical shock or impact;
- › Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- › Pay attention to all factors that may affect the performance of the ranging module;

- After the laser rangefinder module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

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01

Product Overview

This laser ranging module is based on the Time-of-Flight (ToF) principle and supports a measuring range of 3 – 3000 m. It features a UART (TTL 3.3 V) communication interface and supports functions such as single-shot ranging and continuous ranging. The product complies with EU RoHS requirements.

The module utilizes a 905 nm Class 1 eye-safe laser and features a compact design, making it ideal for integration into civilian applications such as UAVs, outdoor observation systems, engineering surveying, and mapping equipment.

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Key Features

> Ultra-Long-Range Ranging Capability

Maximum ranging distance ≥ 3000 m (visibility ≥ 6 km, 60% reflectivity building target), with a minimum ranging distance of 3 m.

> Eye-Safe Laser

Complies with IEC 60825-1 Class 1 requirements, ensuring safe and reliable operation.

> High-Accuracy Measurement

Ranging accuracy: $\pm (0.4 \text{ m} + d \times 0.07\%)$ for distances $\leq 800 \text{ m}$, and $\pm 1 \text{ m}$ for distances $> 800 \text{ m}$.

> Flexible Ranging Modes

Supports both single-shot ranging and continuous ranging with adaptive frequencies from 1 to 5 Hz, meeting the requirements of various applications.

> Low Power Consumption

Standby power consumption is as low as 0.5 mW (POWER_ON pulled low), while operating power consumption is $\leq 0.8 \text{ W}$ at 25°C.

> Standard UART Communication

Features a TTL 3.3 V UART interface with adjustable baud rates (115200 bps by default), offering a simple protocol and easy system integration.

> Fast Startup

Startup time is $\leq 200 \text{ ms}$, enabling rapid system response.

> Excellent Environmental Adaptability

Designed for demanding environments, with an operating temperature range of -20°C to +70°C, shock resistance up to 1200 g, vibration resistance up to 2.5 g, and an MTBF of ≥ 1500 hours.

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Key Specifications

Category	Parameter	Specification
Electrical Specifications	Maximum Ranging Distance (Visibility ≥ 6 km)	≥ 3000 m @ 60% reflectivity, building target ≥ 1800 m @ 30% reflectivity, 2.3×2.3 m target ≥ 1150 m @ 30% reflectivity, 0.5×1.7 m target ≥ 700 m @ 30% reflectivity, 0.2×0.3 m target
	Minimum Ranging Distance	3 m
	Ranging Accuracy (Building Target)	$d \leq 800$ m: $\pm(0.4 \text{ m} + d \times 0.07\%)$ m $d > 800$ m: ± 1 m
	Measurement Frequency	1–5 Hz (Adaptive)
	Measurement Success Rate	$\geq 98\%$
	False Alarm Rate	$\leq 1\%$
	Interface Type	UART (TTL 3.3 V / 2.5 V Compatible)
	Supply Voltage	DC 3–5 V
	Standby Power Consumption	≤ 0.5 mW (POWER_ON)

Category	Parameter	Specification
		Pulled Low) ≤0.3 W (POWER_ON Pulled High)
	Operating Power Consumption (+5 V Supply)	≤0.8 W @25°C ≤0.88 W @70°C ≤0.8 W @-20°C
	Startup Time	≤200 ms
Optical Specifications	Eye Safety Classification	Class 1 (IEC 60825-1)
	Laser Wavelength	905±10 nm
	Laser Beam Divergence (Typical)	2×3.5 mrad
Physical Characteristics	Weight	20±0.5 g
	Dimensions (L × W × H)	34×28×20 mm
	Shock Resistance	1200 g, 1 ms
	Vibration Resistance	5–50–5 Hz, 1 Octave/min, 2.5 g
Environmental Specifications	Operating Temperature	-20°C to +70°C
	Storage Temperature	-25°C to +75°C
	Reliability	MTBF ≥1500 h

Note 1: Visibility refers to visibility under haze conditions. The ranging performance may be affected more significantly under sandstorm conditions.

Note 2: The actual signal high-level voltage is 2.5 V and is compatible with TTL 3.3 V UART interfaces.

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Interface and Connection

Electrical Interface:

Interface Type: UART (TTL 3.3V / 2.5V Compatible)

Connector Model: FWF08002-S06B13W5M (TXGA Connector)

The pin definitions are listed in the table below:

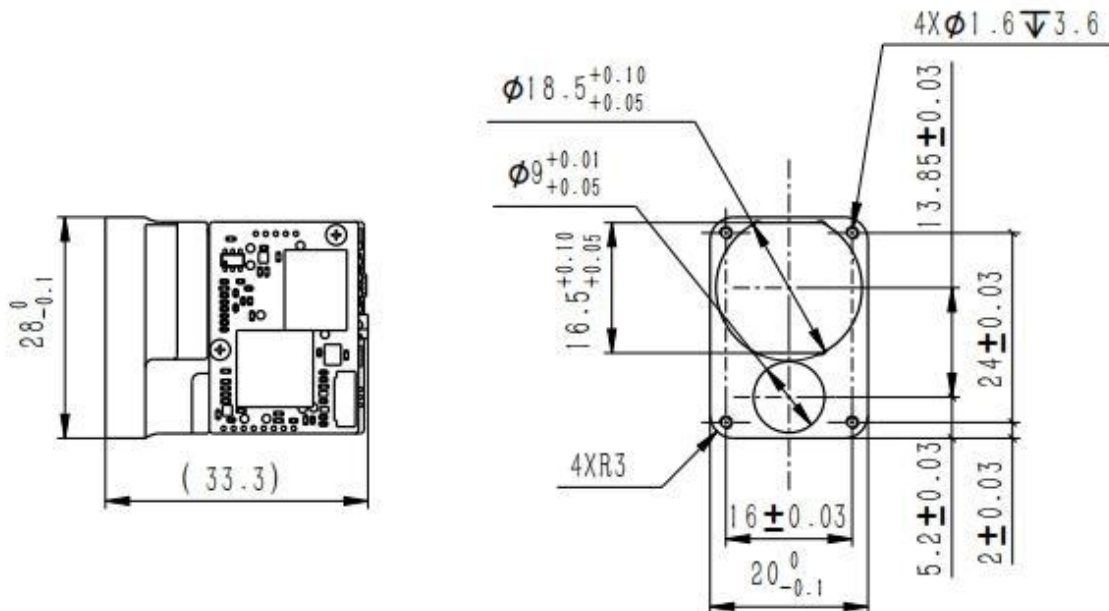
Pin No.	Label	Electrical Definition	Remarks
1	POWER_ON	Module Power Control, TTL 3.3V Level	Module ON (>2.7V), Module OFF (<0.3V)
2	UART_RX	UART Receive, TTL 3.3V Level	Host → Module
3	UART_TX	UART Transmit, TTL 3.3V Level	Module → Host
4	NC	No Connection	-
5	VCC	Power Supply, 3–5V	-
6	GND	Signal Ground	-

Cable Definition (Standard Cable Wiring):

CN1 (C-WB0812H-5-1-S-N)	CN2 (C-WB0812H-6-1-S-N)	Wire Specification	Color
1	-	32AWG	Black
2	-	32AWG	Red
3	-	32AWG	Yellow
4	-	32AWG	Green
5	-	32AWG	White

Mechanical Interface and Center of Gravity Location:

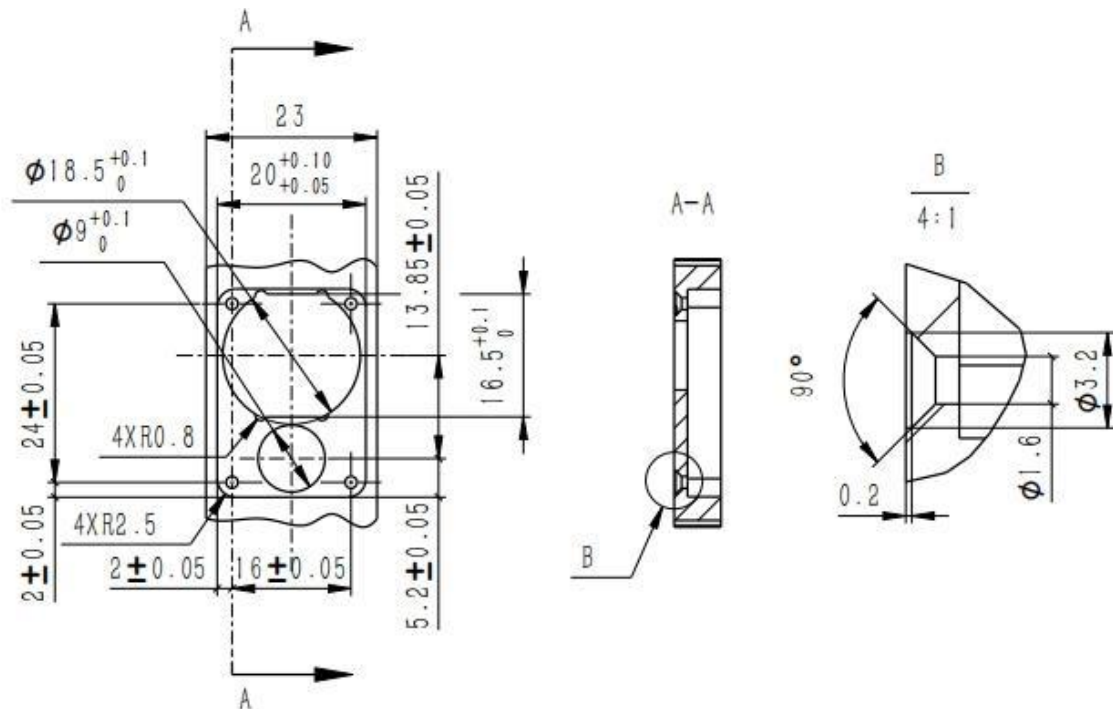
The overall dimensions and center of gravity location are shown in the figure below (unit: mm).

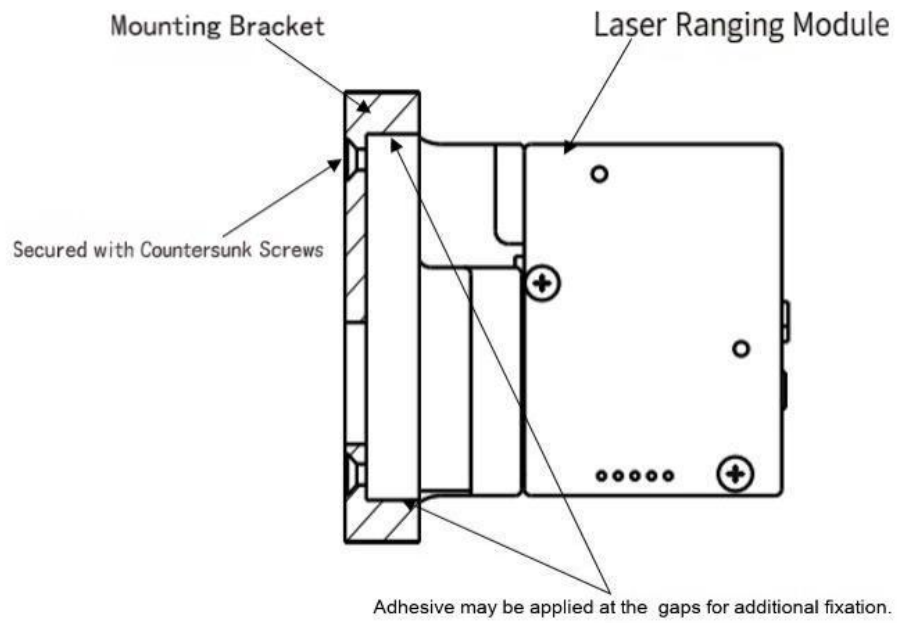


Note: Detailed 3D envelope dimensions shall be subject to the drawings mutually approved by both parties.

Mechanical Installation Notes:

The recommended mounting bracket dimensions and installation diagram are shown below. The thread engagement length of the countersunk screws shall not exceed 3.5 mm.





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Communication Protocol and Commands

For details of the following contents, please refer to Appendix 1:

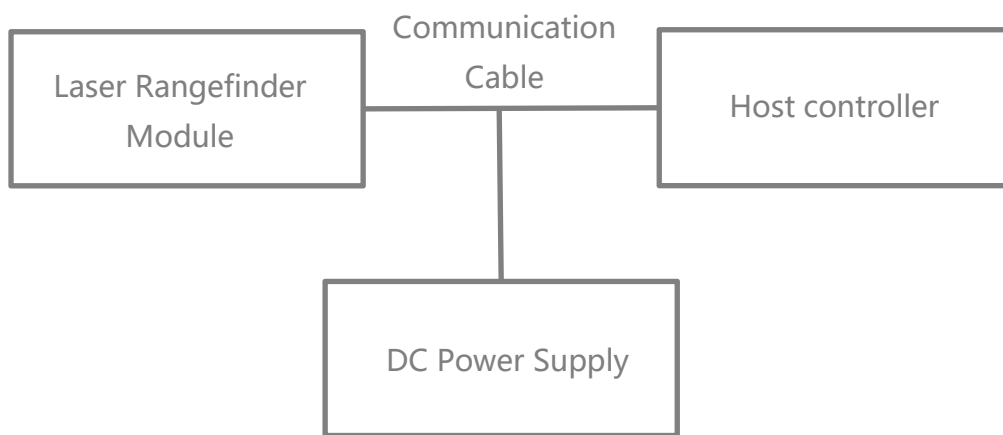
- Protocol Format (Baud Rate, Data Format, and Packet Structure)
- Ranging Commands (Device Self-Test, Single Ranging, Continuous Ranging, and Stop Ranging)
- Configuration Commands (Baud Rate Setting)
- Other Quick Commands (Laser Spot Brightness Adjustment, Firmware Version Query, and Device SN Query)

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Operating Instructions

Power-On Procedure

- Before powering on, connect the laser rangefinder module, communication cable, DC power supply, and host computer according to the electrical interface requirements and as shown in the figure below.



- Ensure that the POWER_ON pin is initially at a low logic level (default state). Pull the POWER_ON pin high (>2.7V) and wait approximately 200 ms (to allow the internal capacitors to fully charge) before sending any commands. This prevents the module from becoming unresponsive.
- The module powers up once the supply voltage is applied. If the POWER_ON pin remains at a high logic level during standby and is not pulled low, subsequent operations can be performed immediately without waiting an additional 200 ms.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the application has exited properly.

- Pull the POWER_ON pin low ($<0.3V$) to place the module into low-power standby mode or power-off state.
- After confirming the above steps, disconnect the power supply to complete a normal shutdown.

Ranging Mode Operation

- Send the "Single Ranging" command to the laser ranging module. The module will perform a single measurement and return the ranging status and measured distance.
- Send the "Continuous Ranging" command to the laser ranging module to start continuous distance measurement.
- Send the "Stop Ranging" command to terminate the ranging operation.

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Precautions for Use

- Do not look directly into the laser emission aperture. Although the laser complies with Class 1 eye-safety requirements, unnecessary exposure should still be avoided.
- Avoid operating the module in environments with excessive heat, humidity, rain, snow, or dust. The operating temperature shall be maintained within -20°C to +70°C.
- Do not disassemble the module. Unauthorized disassembly may void the warranty and result in product damage.
- This product is intended for civilian applications only and must not be used for military purposes.
- The laser spot brightness adjustment mode is intended for development and debugging purposes only. Laser output power is increased in this mode; therefore, the laser beam must not be directed toward human eyes.
- When installing the module, the thread engagement length of countersunk screws shall not exceed 3.5 mm to prevent damage to internal components.
- The supply voltage shall be within the range of DC 3–5 V. Ensure correct power polarity before operation.
- Avoid excessive mechanical shock and vibration. Exposure beyond the specified limits may degrade product performance.

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Optical Window Requirements

- If a protective window is required during system integration, it is recommended that the window provide a transmittance of $\geq 98\%$ within the 905 ± 20 nm wavelength range to minimize optical power loss and ensure optimal ranging performance.

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Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFA3000B laser rangefinder module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

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Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Power Cable	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;

- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure

- Check whether the product appearance is normal;
- Check whether the cable connections are correct and securely connected.

- Power-On Inspection Procedure

- Perform the power-on procedure and verify that the standby power is correct;
- Start the self-test function of the product;
- After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the laser rangefinder module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)

- All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
- All screws on the panel shall be securely fastened;
- The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.

- Comprehensive Power-On Inspection and Maintenance

- Connect the product power supply in sequence;
- Perform the power-on procedure and verify that the standby power is correct;
- Start the product self-test and complete the self-test procedure;
- Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem Symptom	Possible Cause	Troubleshooting Method	Diagnosis Criteria	Solution
Measured distance is shorter than the actual distance	Testing performed under heavy haze, fog, or mist conditions	Test again under favorable weather conditions	If ranging performance returns to normal after weather conditions improve, the issue is determined to be environmental	Avoid operation in adverse weather conditions
Reduced ranging performance, unable to achieve the specified range	① Target reflectivity is extremely low (e.g., water surfaces, wet roads, glass); ② Laser beam is not perpendicular to the target surface; ③ Direct exposure to sunlight	Replace the target with a highly reflective object (e.g., building wall), adjust the aiming angle, and avoid direct strong light	If ranging performance recovers after changing the target or adjusting the angle, the issue is determined to be caused by environmental or operational factors	Select appropriate targets, maintain near-perpendicular incidence, and avoid backlighting conditions

No response when communicating with a PC using the supplied communication cable	Communication cable damaged or broken (the 6-pin cable is thin and susceptible to breakage)	Inspect the cable visually and verify continuity with a multimeter	If communication returns to normal after replacing the cable, the fault is determined to be a cable issue	Replace the communication cable
No response when communicating using a user-supplied cable	Incorrect wiring sequence, short circuit, open circuit, poor contact, or reversed TX/RX connections	Verify the wiring sequence and reconnect according to the pin definition table	If communication returns to normal after rewiring, the fault is determined to be a wiring issue	Reconnect according to the correct pin assignment

Packaging and Transportation Requirements

> Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

> Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

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Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

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Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District,
Chengdu, Sichuan, China

Appendix 1:

1. Protocol Format

Baud Rate: 115200 bps (default) / 57600 bps / 38400 bps / 9600 bps

Data Format: 1 Start Bit, 8 Data Bits, 1 Stop Bit, No Parity

Packet Format (Field Length: 1 Byte, Hexadecimal):

Field	Length	Value Range	Description
Header	2	0xEE 0x16	Fixed Bytes
Data Length	1	2~7	Number of Following Valid Bytes
Device Code	1	0x03	Fixed Byte
Command Code	1	0x00~0xFF	Command Type Identifier
Parameters	0~5	0x00~0xFF	Parameter Values (Depending on Command)
Checksum	1	0x00~0xFF	Sum of Device Code + Command Code + Parameters (Lower 8 Bits)

2. Ranging Commands

- **All commands below are transmitted in hexadecimal format.**
- **Device Self-Test**

Transmit: EE 16 02 03 01 04

Response: EE 16 06 03 01 Status3 Status2 Status1 Status0 Check

SStatus3, Status2: Reserved

SStatus1: bit0 – FPGA System Status (1 = Normal, 0 = Fault); bit5 – Bias Voltage Output Status (1 = Normal, 0 = Fault); bit6 – Temperature Status (1 = Normal, 0 = Fault)

SStatus0: Reserved

> **Single Ranging**

Transmit: EE 16 02 03 02 05

Response: EE 16 06 03 02 Status Result2 Result1 Result0 Check

Status: 0x00 = Valid Single Target Measurement; 0x04 = Out of Range

Result2 (High Byte), Result1 (Low Byte), Result0 (Decimal Digit)

Distance = Result2 × 256 + Result1 + Result0 × 0.1 (Unit: m)

> **Continuous Ranging**

Transmit: EE 16 02 03 04 07

Response: EE 16 06 03 04 Status Result2 Result1 Result0 Check

Data parsing method is identical to Single Ranging.

> **Stop Ranging**

Transmit: EE 16 02 03 05 08

Response: EE 16 02 03 05 08 (Echoes the transmitted command)

3. Configuration Commands

> Set Baud Rate

Transmit: EE 16 06 03 A0 BD24 BD16 BD8 BD0 Check

Response: Same as transmitted command

The baud rate is a 32-bit unsigned integer (U32), transmitted with the most significant byte first (BD24~BD0). The value represents the actual baud rate (e.g., 115200).

4. Other Quick Commands

> Enter Laser Spot Brightness Adjustment Mode

Transmit: EE 16 04 03 10 01 power Check (power = 1–100%, brightness level)

Response: Same as transmitted command

This command is intended solely for laser spot brightness debugging and must not be used during normal ranging operation.

> Exit Laser Spot Brightness Adjustment Mode (Return to Normal Ranging Mode)

Transmit: EE 16 04 03 10 00 00 13

Response: Same as transmitted command

> **Query FPGA Version**

Transmit: EE 16 02 03 A6 A9

Response: EE 16 06 03 A6 Version1 Version2 Version3 Version4
Check

> **Query MCU Version**

Transmit: EE 16 02 03 A7 AA

Response: EE 16 06 03 A7 Version1 Version2 Version3 Version4
Check

> **Query Hardware Version**

Transmit: EE 16 02 03 A8 AB

Response: EE 16 06 03 A8 Version1 Version2 Version3 Version4
Check

> **Query Device Serial Number (SN)**

Transmit: EE 16 02 03 A9 AC

Response: EE 16 06 03 A9 SN1 SN2 SN3 SN4 Check